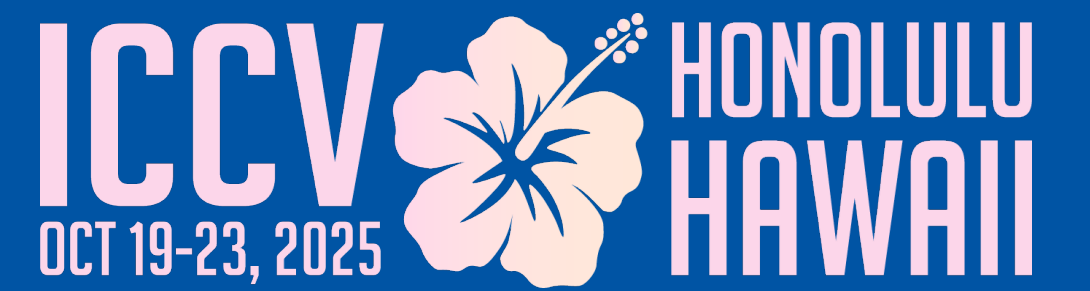




ProGait: A Multi-Purpose Video Dataset and Benchmark for Transfemoral Prosthesis Users

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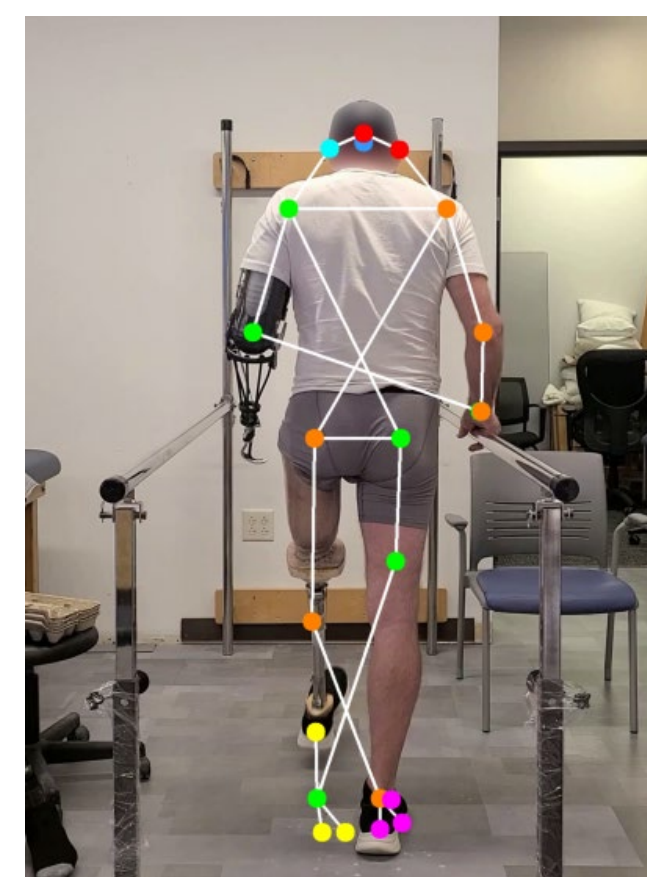
Motivation

Background: Vision-based ML approaches provide a promising pathway to prosthetic gait analysis, which is important for prosthesis design and alignment.

Challenge: Current vision models face significant limitations in detecting prosthetic legs as part of the human body.

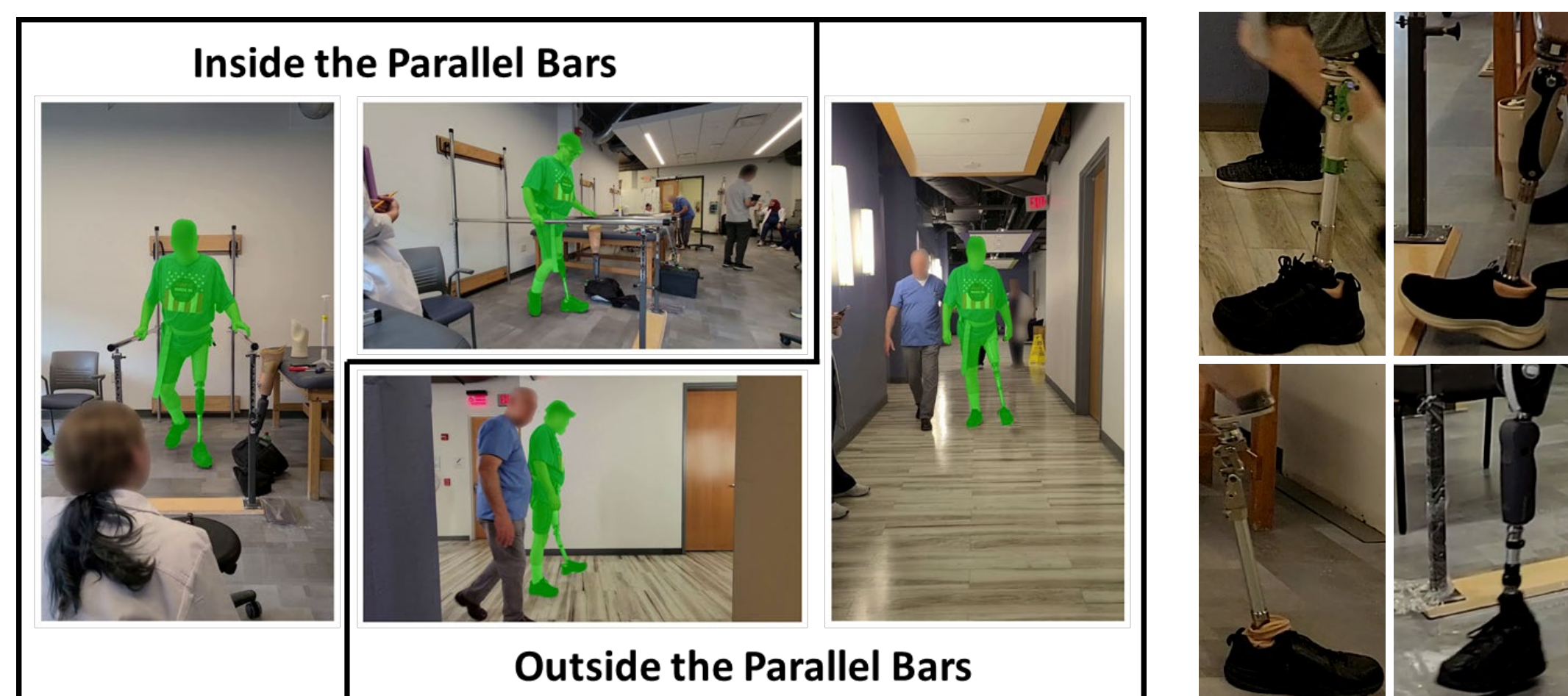


Segmentation (YOLO11)



Pose Estimation (RTMW)

Data Collection



Left: The recording scenarios. Right: Different types of prosthetic legs.

Contribution

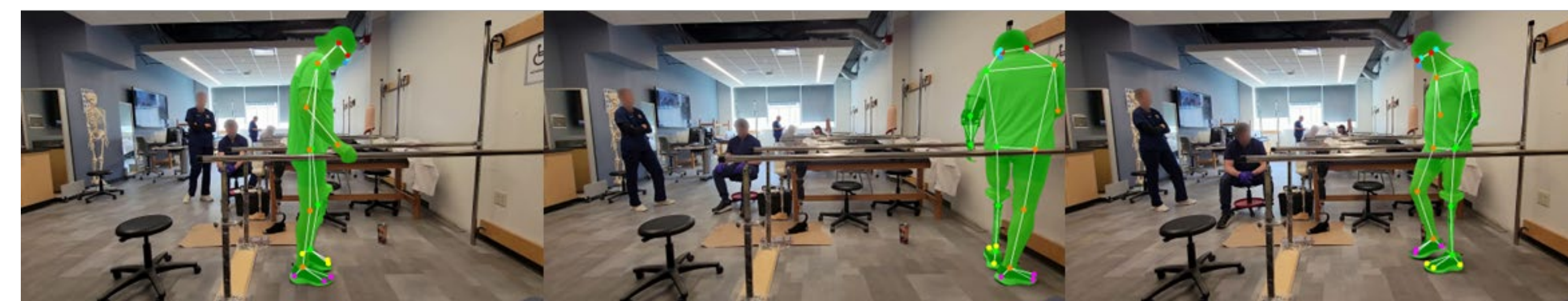
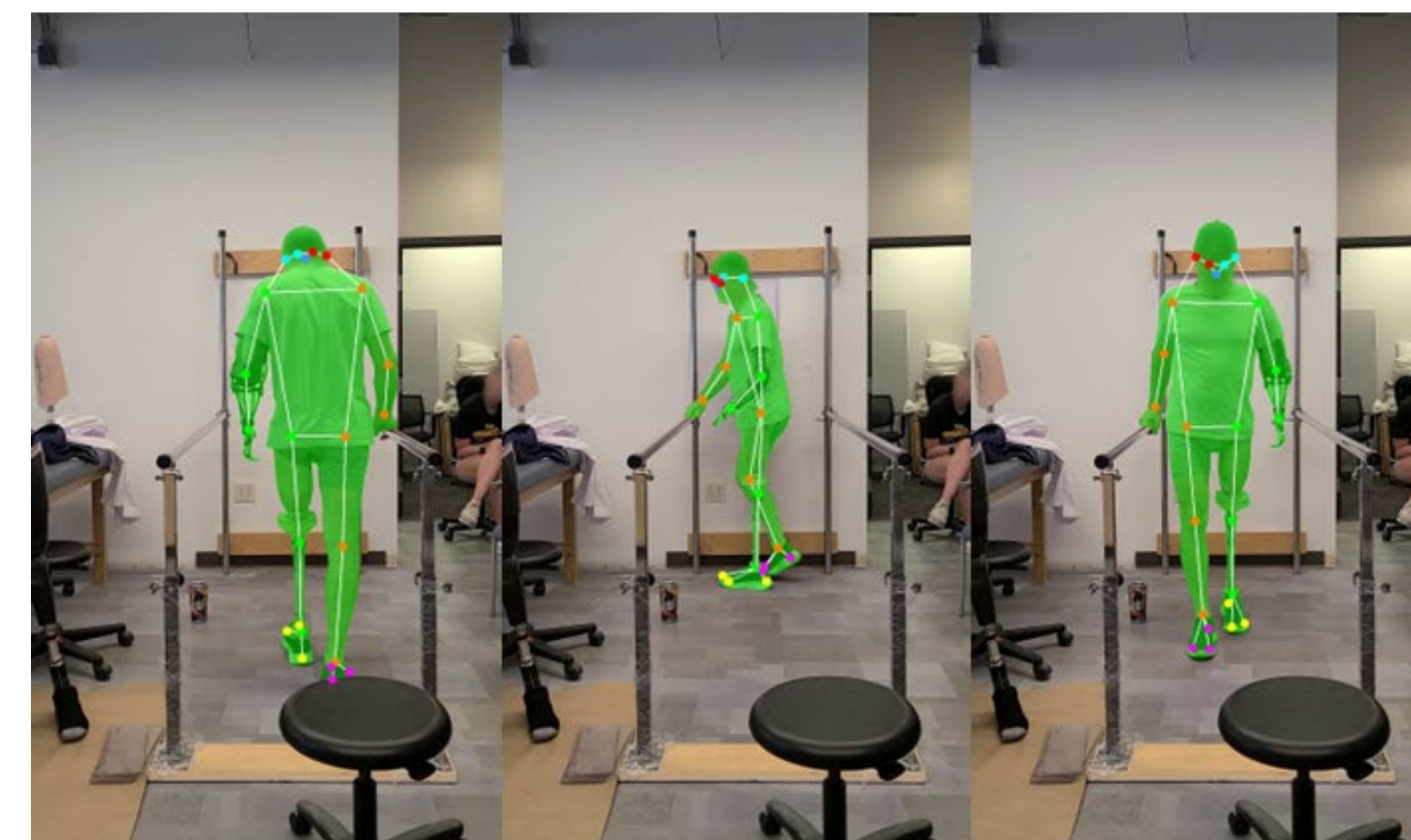
ProGait dataset & benchmark →

Tasks:

- Video Object Segmentation
- 2D Human Pose Estimation
- Gait Analysis

Data Volume:

- 412 video clips
- 4 above-knee amputees
- Multiple newly-fitted prosthetic legs per subject
- 2 scenarios – inside and outside the parallel bars.
- 9 categories of prosthetic gait



Top-left: Frontal view. Bottom: Sagittal view. Top-right: Textual descriptions for gait analysis.

Coarse Gait Category:
3. Step Length and Timing Issues

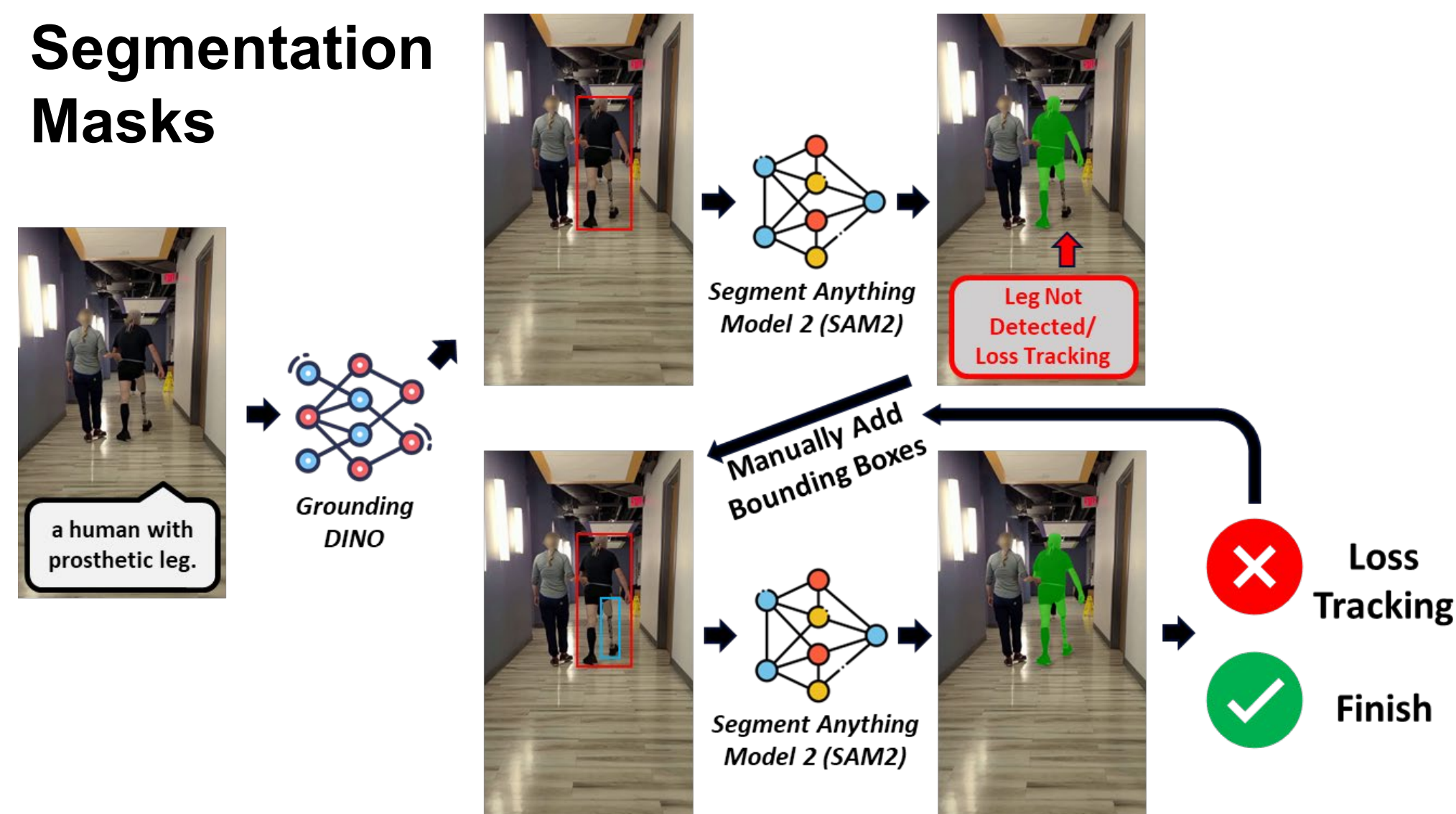
Gait Deviation:
Terminal impact

How to correct:
Reduce swing flexion resistance/decrease extension assist tension

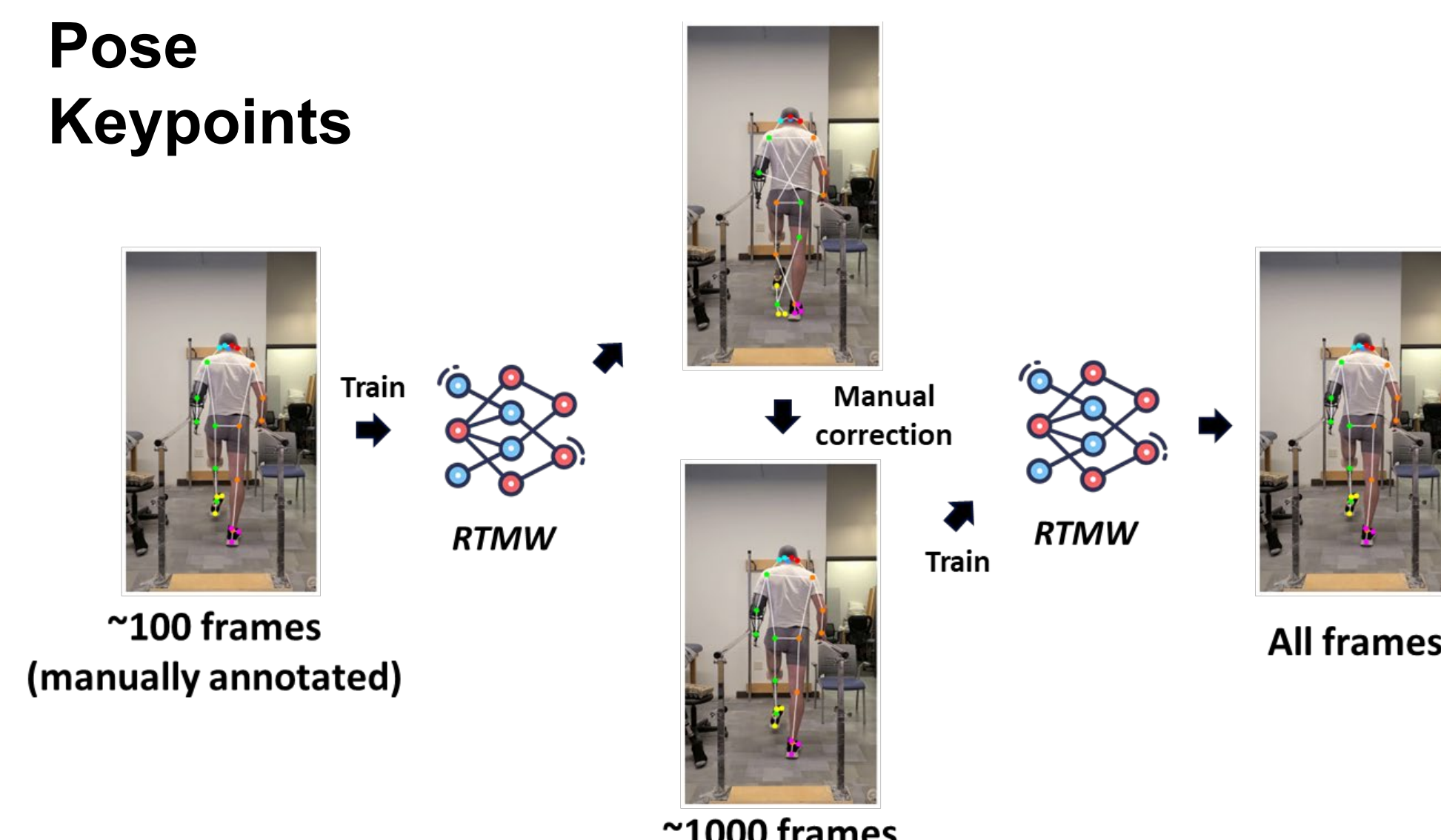
Reason:
The knee needs to flex more during swing phase to improve toe clearance and to better synchronize full extension with heel strike. This is best optimized at normal gait speed (not in the parallel bars)

Annotation

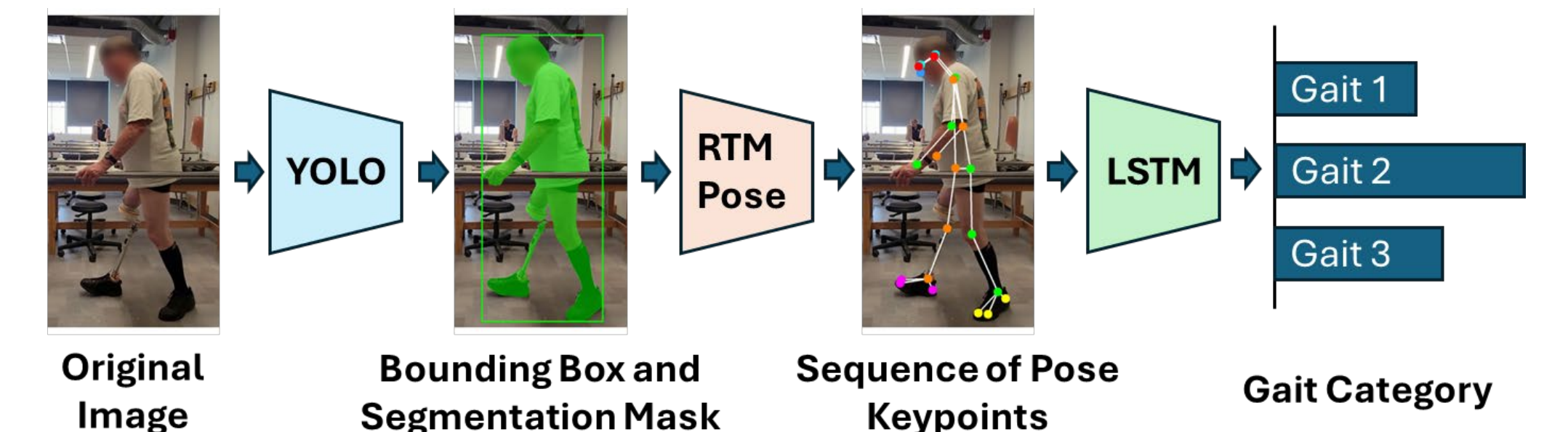
Segmentation Masks



Pose Keypoints



Baseline Models & Evaluations



Video Object Segmentation

Method	mIoU	mIoU (inside)	mIoU (outside)
YOLO11	0.784	0.831	0.774
Grounded SAM2			
“a person.”	0.358	0.643	0.559
“an amputee.”	0.905	0.90	0.907
“a human with prosthetic leg.”	0.964	0.921	0.929
YOLO11-ProGait	0.847	0.815	0.866

2D Human Pose Estimation

Method	AP	AP (inside)	AP (outside)
HRNet [35]	0.750	0.825	0.733
ViPNAS [37]	0.761	0.735	0.767
RTMPose [18]	0.855	0.876	0.850
ViTPose [38]	0.830	0.845	0.822
RTMPose-ProGait	0.947	0.968	0.942

Gait Classification (all keypoints)

	Top-1 Acc.	Balanced Acc.
Frontal view	0.510	0.545
Sagittal view	0.826	0.790
Inside parallel bars	0.364	0.437
Outside parallel bars	0.486	0.320
Whole dataset	0.372	0.403

Gait Classification (lower-body only)

	Top-1 Acc.	Balanced Acc.
Frontal view	0.474	0.521
Sagittal view	0.773	0.812
Inside parallel bars	0.364	0.457
Outside parallel bars	0.566	0.423
Whole dataset	0.384	0.413

Gait Classification (vs. others)

	Pre-trained	Fine-tuned/ Re-trained
GaitGraph2 [36]	0.200	0.440
GPGait [12]	0.388	0.457
GaitBase [11]	0.313	0.340
ScoNet	N/A	0.333
LSTM-ProGait	N/A	0.372

Conclusion

ProGait effectively improves the generalizability of vision models for individuals with prosthetic legs by providing high-quality video data, thereby bridging the gap between computer vision research and real-world clinical needs.